

IMPLICATIONS AND RELATED WORKS

- Understanding LWE and SIS quantumly $\rightarrow$ find better attacks on SIS.
 - ▶ [BKSW19, CLZ21]: studied reductions between LWE, SIS and related quantum problems, and defined quantum versions of LWE and SIS;
 - ▶ [BJKNY25]: improved existing attacks on one of these quantum versions;
 - ▶ [CH26]: provided a reverse reduction $\text{SIS} \leftarrow \text{LWE}$.
- Regev's reduction with (classical) error correcting codes.
 - ▶ [CT23, CT24, BC25, BCT26] defined Regev's reduction for codes, then applied it to different types of codes and metrics to improve attacks;
 - ▶ [Google team 2025] Regev's reduction with specific codes and noise $\rightarrow$ solve an optimization problem (variant of SAT);
 - ▶ [YZ22] Regev's reduction with specific codes and noise $\rightarrow$ proved quantum advantage relatively to a random oracle only.
- Using complex amplitudes.
 - ▶ [DFS24] used a Regev's style strategy with quantum amplitudes to build an *LWE oblivious sampler* and break a conjecture;
 - ▶ [CHLLT] studied Regev's reduction with quantum amplitudes in general.